Chapter 1. Abstract Integration.

1. Does there exist an infinite σ-algebra which has only countably many members?

Proof. Let $\mathcal{M}$ be a σ-algebra on a set X.
Suppose that $\mathcal{M}$ has only countably infinite members.
If X is finite set, then the power set $\mathcal{P}(X)$ is finite, so is $\mathcal{M} \subset \mathcal{P}(X)$.
Therefore, X must be infinite if $\mathcal{M}$ is infinite.
Since $\mathcal{M}$ is countably infinite, enumerate $\mathcal{M} = \{A_i \mid i \in \mathbb{N}\}$.
For each $x \in X$, define $B_x := \bigcap_{\substack{x \in A, \\ A \in \mathcal{M}}} A$. Then, B_x is a countable intersection of $A_i \in \mathcal{M}$, so $B_x \in \mathcal{M}$.

Assume that: for any $x, y \in X$, $B_x \cap B_y \neq \emptyset$. Let $z \in B_x \cap B_y$.
Then, for all $A \in \mathcal{M}$ containing x, $z \in A$. So,

$$B_z = \bigcap \{A \mid A \in \mathcal{M},\ z \in A\} \subseteq \bigcap \{A' \mid A' \in \mathcal{M},\ x \in A'\} = B_x$$

and similarly $B_z \subseteq B_y$, thus $z \in B_z \subseteq B_x \cap B_y$.
Meanwhile, assume that $x \notin B_z$. Then, $x \in B_x \setminus B_z$. But, since $B_x, B_z \in \mathcal{M}$, $B_x \setminus B_z = B_x \cap B_z^c \in \mathcal{M}$, so that $B_x \setminus B_z = B_x$. It contradicts to $z \in B_x$.
So, $x \in B_z$ and similarly $y \in B_z$. Therefore $B_z = B_x \cap B_y = B_x = B_y$.
Let $\mathcal{B} := \{B_x \mid x \in X\}$. Assume $\mathcal{B}$ is finite. Notice that: given $A \in \mathcal{M}$,

$$A = \bigcup_{x \in A} B_x, \text{ so that } \quad \#\mathcal{M} \leq \#\{0,1\}^{\mathcal{B}} < \aleph_0,$$

but it is contradition, since $\#\mathcal{M} = \aleph_0$.
Assume $\mathcal{B}$ is uncountably infinite. Then, since $\mathcal{B} \subseteq \mathcal{M}$, $\mathcal{M}$ is also uncountable, so it contradicts.
Now, suppose $\mathcal{B}$ is countable, and let enumerate $\mathcal{B} = \{B_i \mid i \in \mathbb{N}\}$ which satisfies that $B_i = B_j \Leftrightarrow i = j$ and $i \neq j \Rightarrow B_i \cap B_j$.
Then, for any $F_1, F_2 \subseteq \mathbb{N}$, $F_1 \neq F_2$ implies that

$$\bigcup_{i \in F_1} B_i \neq \bigcup_{j \in F_2} B_j, \text{ and } \bigcup_{n \in F} B_n \in \mathcal{M} \text{ for each } F \subseteq \mathbb{N}.$$

Now, a map

$$\mathbb{N} \to \mathcal{M} : F \mapsto \bigcup_{n \in F} B_n$$

is well-defined injective map.
Therefore, $\#\mathcal{P}(\mathbb{N}) = 2^{\aleph_0} = \#\mathcal{M}$, it contradicts, so $\mathcal{M}$ is uncountable. ∎

Similar question
Does there exists an infinite topology which has only countably many members?

Yes. Let $X=\mathbb{N}$ with cofinite topology.
But, there is a fun fact:

Infinite Hausdorff topology must be uncountable.

3. Prove that: if f is a real function on a measurable space X such that $\{x \in X \mid f(x) \geq r\}$ is measurable for all $r \in \mathbb{R}$, then f is measurable.

Proof.